Adam Scholey

Software Engineer

Core: Python · FastAPI · PostgreSQL · Docker · CI/CD · Kubernetes · OWASP · SAST/DAST
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Personal Statement

A backend-focused Software Engineer with hands-on experience building secure, production-oriented systems, primarily using Python and PostgreSQL. I specialise in building multi-tenant architectures with a strong emphasis on data security, including an enterprise search engine with row-level security and GDPR-compliant cross-border data access, and a live client platform built on FastAPI with Stripe payments and secure authentication. I achieved Distinction-level results in the cyber security units of my BTEC IT (D*D*D) and bring that security mindset into every system I build, from access control to data isolation. I am seeking a backend or full-stack software engineering role where I can apply this foundation to real-world products in a professional environment.

Experience

Software Engineer (Part-time)

diam Systems Ltd · Oct 2025 - Present

- Built a multi-tenant enterprise search engine using Python and PostgreSQL, with GDPR-compliant cross-border data access.
- Implemented advanced PostgreSQL architecture with tsvector full-text search, row-level security, and table partitioning.
- Developed an internal project management tool in active daily use across the team, with real-time SSE updates and CLI integration.
- Built a production-ready restaurant booking system using Python and Flask, with Stripe payments and a visual floor plan.
- Building a paid client platform (framed@) on FastAPI with async SQLAlchemy, PostgreSQL, Stripe checkout, and passwordless authentication.
- Contributed to diamsystems.co.uk, the company website with e-commerce (Printify + Stripe) and i18n support.

Production Operator (Part-time)

Pure Pet Food Ltd · Oct 2024 - Present

- Operated production machinery in a fast-paced manufacturing environment.
- Maintained quality standards and met daily output targets.

Prototype Tester

Microsoft, Cambridge · 2019

- Selected by Microsoft to test an accessibility-focused mobile application prototype as a partially sighted user.
- Provided structured feedback on screen reader compatibility, contrast, and UX for visually impaired users.

Farm Hand

Brians Best Wools · Sept 2023 - Aug 2024

- Managed daily production tasks to strict schedules in a physically demanding environment, demonstrating reliability and consistency.

Skills

Technical Skills

BACKEND

- FastAPI - async endpoints, Pydantic, webhooks
- ASP.NET Core - REST APIs, middleware, auth pipelines
- Flask - async routes, blueprints, SQLAlchemy
- SignalR - real-time hubs and SSE
- OAuth 2.0 / GitHub OAuth with Authlib
- Rate limiting, JWT, BCrypt, RBAC

DATABASES

- PostgreSQL - RLS, migrations, query optimisation
- SQLite - lightweight production DBs
- EF Core - code-first schema, relationships
- SQLAlchemy - ORM patterns and raw queries
- Redis - caching, pub/sub, distributed locks

SECURITY

- SAST / DAST / SCA - Semgrep, Snyk, GitLeaks, pip-audit
- OWASP ZAP - authenticated security testing
- AES-256 encryption, key rotation design
- Zero-knowledge encryption patterns
- GDPR compliance, CSP, input validation

TESTING

- xUnit - unit and integration tests
- pytest - fixtures, parametrize, coverage
- k6 - load testing and performance baselines
- Playwright - E2E browser automation
- 370+ tests across production systems - unit, integration and E2E (158 on framed@ alone)

DESKTOP & MOBILE

- WinUI 3 - modern Windows applications
- .NET MAUI - cross-platform desktop/mobile
- Blazor - component-based web UIs
- CommunityToolkit.Mvvm - MVVM patterns
- PWAs - progressive web apps

INFRASTRUCTURE

- Docker - containerisation and Compose
- AWS S3 - object storage and signed URLs
- Git, GitHub Actions, CI/CD pipelines
- Cloudflare and Cloudflare Tunnel
- Kubernetes - Deployment, HPA, NetworkPolicy

Projects

framed@ · Client Booking & Events Platform

FastAPI, PostgreSQL, Stripe · framedat.uk

- Paid client engagement building a music and events platform for framed@ Limited - scoped features, estimated effort, demoed progress and delivered against agreed milestones.
- FastAPI backend with async SQLAlchemy, PostgreSQL, and Pydantic; Stripe hosted checkout and webhooks.
- Passwordless magic-link authentication and real-time admin dashboard via SSE.
- 158 automated tests across unit, route-level and security-regression suites; live preview deployed ahead of production launch.

Enterprise Search Platform

Python, Flask, PostgreSQL, Redis, Kubernetes

- Multi-tenant SaaS platform with multi-region deployment across UK and EU.
- Advanced PostgreSQL architecture: tsvector full-text search, pg_trgm GIN indexes, row-level security, table partitioning by region, and PgBouncer connection pooling.
- GDPR-compliant cross-border data access with consent management, audit logging, and data sovereignty enforcement across UK and EU regions.
- Kubernetes deployment with horizontal pod autoscaling, multi-connector parallel search via asyncio, and guided database connection wizard with SSH tunnel.

Internal Project Management Tool

Python, Flask, SQLite

- Elegant project management tool with seven-stage lifecycle tracking.
- Real-time updates via Server-Sent Events (SSE) and email notifications.
- CLI package for command-line project management with REST API integration.
- Deployed to Fly.io with SQLite database and SMTP email notifications.

Zero-Knowledge Cloud Storage Platform

C#, Blazor, ASP.NET Core, PostgreSQL, AWS S3, SignalR

- Client-side AES-256 encryption before upload, so the server stores ciphertext only and never sees plaintext.
- Real-time collaborative editing via SignalR with Monaco editor and operational-transform conflict handling.
- AWS S3 object storage integration with signed URL handling for uploads and downloads.

Education

Bradford College

Sept 2023 - Jun 2026

- Level 1/2 Information and Creative Technology - DM
- English Language (6), Mathematics (4)
- BTEC Level 3 National Extended Diploma in Information Technology - D*D*D
- **Award:** Learner of the Year, Computing, Science & Environmental Technology, University of Bradford · Skillshouse CTE.

Trinity Academy Bradford

Sept 2018 - June 2023

- English Literature (5)
- BTEC Level 1/2 in Sport (Distinction)
- BTEC Level 1/2 in Engineering (Merit)
- BTEC Level 1/2 in Creative iMedia (Merit)